

MODULE 7 AND 8

1. Try Test-Connection and nslookup commands for below websites

www.google.com

www.facebook.com

www.amazon.com

www.github.com

Test-Connection does dns resolution first and does an icmp ping.

```
Windows PowerShell
PS C:\Users\yasha> Test-Connection www.google.com

Source      Destination    IPV4Address    IPV6Address
-----      -
LAPTOP-5LJ... www.google.com 142.251.153.119 2404:6800:4002:818::2004
LAPTOP-5LJ... www.google.com 142.251.153.119 2404:6800:4002:818::2004
LAPTOP-5LJ... www.google.com 142.251.153.119 2404:6800:4002:818::2004
LAPTOP-5LJ... www.google.com 142.251.153.119 2404:6800:4002:818::2004

PS C:\Users\yasha> Test-Connection www.facebook.com

Source      Destination    IPV4Address    IPV6Address
-----      -
LAPTOP-5LJ... www.facebook... 57.144.56.1     2a03:2880:f335:1:face:b00...
LAPTOP-5LJ... www.facebook... 57.144.56.1     2a03:2880:f335:1:face:b00...
LAPTOP-5LJ... www.facebook... 57.144.56.1     2a03:2880:f335:1:face:b00...
LAPTOP-5LJ... www.facebook... 57.144.56.1     2a03:2880:f335:1:face:b00...

PS C:\Users\yasha> Test-Connection www.amazon.com

Source      Destination    IPV4Address    IPV6Address
-----      -
LAPTOP-5LJ... www.amazon.com 18.67.150.142   2600:140f:400:180::3bd4
LAPTOP-5LJ... www.amazon.com 18.67.150.142   2600:140f:400:180::3bd4
LAPTOP-5LJ... www.amazon.com 18.67.150.142   2600:9000:24d8:da00:7:49a...
LAPTOP-5LJ... www.amazon.com 18.67.150.142   2600:9000:24d8:da00:7:49a...

PS C:\Users\yasha> Test-Connection www.github.com

Source      Destination    IPV4Address    IPV6Address
-----      -
LAPTOP-5LJ... www.github.com 20.207.73.82    64:ff9b:14cf:4952
LAPTOP-5LJ... www.github.com 20.207.73.82    64:ff9b:14cf:4952
LAPTOP-5LJ... www.github.com 20.207.73.82    64:ff9b:14cf:4952
LAPTOP-5LJ... www.github.com 20.207.73.82    64:ff9b:14cf:4952
```

```
PS C:\Users\yasha> Test-Connection www.cisco.com

Source      Destination    IPV4Address    IPV6Address
-----      -
LAPTOP-5LJ... www.cisco.com 23.200.49.43    2600:140f:400:18e::b33
LAPTOP-5LJ... www.cisco.com 23.200.49.43    2600:140f:400:186::b33
LAPTOP-5LJ... www.cisco.com 23.200.49.43    2600:140f:400:186::b33
LAPTOP-5LJ... www.cisco.com 23.200.49.43    2600:140f:400:186::b33
```

Nslookup

```
PS C:\Users\yasha> nslookup www.google.com
Server:      UnKnown
Address:     10.13.242.36

Non-authoritative answer:
Name:        www.google.com
Addresses:   2001:4860:4826:7700::
             2001:4860:482c:7700::
             2001:4860:482a:7700::
             2001:4860:4827:7700::
             2001:4860:482d:7700::
             2001:4860:4828:7700::
             2001:4860:4829:7700::
             2001:4860:482b:7700::
             142.251.221.196

PS C:\Users\yasha> nslookup www.facebook.com
Server:      UnKnown
Address:     10.13.242.36

Non-authoritative answer:
Name:        star-mini.c10r.facebook.com
Addresses:   2a03:2880:f334:1:face:b00c:0:25de
             57.144.54.1
Aliases:     www.facebook.com
```

```
Windows PowerShell

PS C:\Users\yasha> nslookup www.facebook.com
Server:      UnKnown
Address:     10.13.242.36

Non-authoritative answer:
Name:        star-mini.c10r.facebook.com
Addresses:   2a03:2880:f334:1:face:b00c:0:25de
             57.144.54.1
Aliases:     www.facebook.com

PS C:\Users\yasha> nslookup www.amazon.com
Server:      UnKnown
Address:     10.13.242.36

Non-authoritative answer:
Name:        cf.47cf2c8c9-frontier.amazon.com
Addresses:   2600:9000:24d9:b000:7:49a5:5fd6:da1
             2600:9000:24d9:9c00:7:49a5:5fd6:da1
             2600:9000:24d9:7a00:7:49a5:5fd6:da1
             2600:9000:24d9:bc00:7:49a5:5fd6:da1
             2600:9000:24d9:de00:7:49a5:5fd6:da1
             2600:9000:24d9:3000:7:49a5:5fd6:da1
             2600:9000:24d9:1c00:7:49a5:5fd6:da1
             2600:9000:24d9:4a00:7:49a5:5fd6:da1
             18.67.150.142
Aliases:     www.amazon.com
             tp.47cf2c8c9-frontier.amazon.com
```

```
PS C:\Users\yasha> nslookup www.github.com
Server:      UnKnown
Address:     10.13.242.36

Non-authoritative answer:
Name:        github.com
Addresses:   64:ff9b::14cf:4952
             20.207.73.82
Aliases:     www.github.com

PS C:\Users\yasha> nslookup www.cisco.com
Server:      UnKnown
Address:     10.13.242.36

Non-authoritative answer:
Name:        e2867.dsca.akamaiedge.net
Addresses:   2600:140f:400:186::b33
             2600:140f:400:181::b33
             23.200.49.43
Aliases:     www.cisco.com
             www.cisco.com.akadns.net
             wwwds.cisco.com.edgekey.net
             wwwds.cisco.com.edgekey.net.globalredir.akadns.net

PS C:\Users\yasha>
```

2. Use Wireshark to capture and analyze DNS, TCP, UDP traffic and packet header, packet flow, options and flags.

DNS

```
Wireshark - Packet 34 - WiFi

> Frame 34: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
> Ethernet II, Src: b0:3c:dc:e3:e5:09, Dst: fe:ea:05:6c:40:99
> Internet Protocol Version 4, Src: 172.28.128.153, Dst: 172.28.128.67
> Transmission Control Protocol, Src Port: 53687, Dst Port: 53, Seq: 3, Ack: 1, Len: 32
> [2 Reassembled TCP Segments (34 bytes): #31(2), #34(32)]
▼ Domain Name System (query)
  Length: 32
  Transaction ID: 0xfe64
  ▼ Flags: 0x0100 Standard query
    0... .. = Response: Message is a query
    .000 0... .. = Opcode: Standard query (0)
    .... .0... .. = Truncated: Message is not truncated
    .... .1... .. = Recursion desired: Do query recursively
    .... ..0... .. = Z: reserved (0)
    .... ..0... .. = Non-authenticated data: Unacceptable
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    > www.google.com: type HTTPS, class IN
    [Response In: 114]
```

```
Wireshark - Packet 114 - WiFi

> Frame 114: Packet, 112 bytes on wire (896 bits), 112 bytes captured (896 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
> Ethernet II, Src: fe:ea:05:6c:40:99, Dst: b0:3c:dc:e3:e5:09
> Internet Protocol Version 4, Src: 172.28.128.67, Dst: 172.28.128.153
> Transmission Control Protocol, Src Port: 53, Dst Port: 53687, Seq: 2, Ack: 35, Len: 58
> [2 Reassembled TCP Segments (59 bytes): #96(1), #114(58)]
▼ Domain Name System (response)
  Length: 57
  Transaction ID: 0xfe64
  ▼ Flags: 0x8180 Standard query response, No error
    1... .. = Response: Message is a response
    .000 0... .. = Opcode: Standard query (0)
    .... .0... .. = Authoritative: Server is not an authority for domain
    .... .0... .. = Truncated: Message is not truncated
    .... .1... .. = Recursion desired: Do query recursively
    .... .1... .. = Recursion available: Server can do recursive queries
    .... ..0... .. = Z: reserved (0)
    .... ..0... .. = Answer authenticated: Answer/authority portion was not authenticated by the server
    .... ..0... .. = Non-authenticated data: Unacceptable
    .... ..0000 = Reply code: No error (0)
  Questions: 1
  Answer RRs: 1
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    > www.google.com: type HTTPS, class IN
  ▼ Answers
    > www.google.com: type HTTPS, class IN
    [Request In: 34]
    [Time: 141.684200 milliseconds]

0000  b0 3c dc e3 e5 09 fe ea 05 6c 40 99 08 00 45 00  <.....l@...E.
0010  00 62 10 00 40 00 06 d1 80 ac 1c 80 43 ac 1c    .b.@.@....C.
0020  80 99 00 35 d1 b7 dd 9b 2e 60 1d fc 5b 37 50 18  ..5....[7P
0030  00 80 bf 3d 00 00 39 fe 64 81 80 00 01 00 01 00  ...=:9:d.....
0040  00 00 00 03 77 77 77 06 67 6f 6f 67 6c 65 03 63  ....www.google.c
0050  6f 6d 00 00 41 00 01 c0 0c 00 41 00 01 00 00 30  om...A...-0
0060  48 00 0d 00 01 00 00 01 00 06 02 68 32 02 68 33  H.....h2-h3

Packet (112 bytes)  Reassembled TCP (59 bytes)

No: 114 - Time: 1.353350300 - Source: 172.28.128.67 - Destination: 172.28.128.153 - Protocol: DNS - Length: 112 - Info: Standard query response 0xfe64 HTTPS www.google.com HTTPS
Show packet bytes  Layout: Vertical (Stacked)
```

TCP

Wireshark - Packet 106 - WiFi

Frame 106: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF...
Ethernet II, Src: fe:ea:05:6c:40:99, Dst: b0:3c:dc:e3:e5:09
Internet Protocol Version 4, Src: 172.28.128.67, Dst: 172.28.128.153
Transmission Control Protocol, Src Port: 53, Dst Port: 61015, Seq: 131, Ack: 45, Len: 0

Source Port: 53
Destination Port: 61015
[Stream index: 2]
[Stream Packet Number: 12]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 131 (relative sequence number)
Sequence Number (raw): 3602032269
[Next Sequence Number: 132 (relative sequence number)]
Acknowledgment Number: 45 (relative ack number)
Acknowledgment number (raw): 2524604028
0101 = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)
Window: 128
[Calculated window size: 65536]
[Window size scaling factor: 512]
Checksum: 0xed79 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
▼ [Timestamps]
[Time since first frame in this TCP stream: 168.947500 milliseconds]
[Time since previous frame in this TCP stream: 3.540200 milliseconds]
> [SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

300 b0 3c dc e3 e5 09 fe ea 05 6c 40 99 08 00 45 00 ..<.....l@...
310 00 28 3a 3d 40 00 00 06 a7 7d ac 1c 80 43 ac 1c ..(:=@@:.)...C..
320 80 99 00 35 ee 57 d6 b2 a6 8d 96 7a 66 7c 50 11 ...5.W...z[P..
330 00 80 ed 79 00 00 ...y...

106 - Time: 1.332230400 - Source: 172.28.128.67 - Destination: 172.28.128.153 - Info: 53 → 61015 [FIN, ACK] Seq=131 Ack=45 Win=65536 Len=0

Show packet bytes Layout: Vertical (Stacked) ▼

Close Help

Wireshark - Packet 107 - WiFi

> Frame 107: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF...
> Ethernet II, Src: b0:3c:dc:e3:e5:09, Dst: fe:ea:05:6c:40:99
> Internet Protocol Version 4, Src: 172.28.128.153, Dst: 172.28.128.67
▼ Transmission Control Protocol, Src Port: 61015, Dst Port: 53, Seq: 45, Ack: 132, Len: 0

Source Port: 61015
Destination Port: 53
[Stream index: 2]
[Stream Packet Number: 13]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 45 (relative sequence number)
Sequence Number (raw): 2524604028
[Next Sequence Number: 45 (relative sequence number)]
Acknowledgment Number: 132 (relative ack number)
Acknowledgment number (raw): 3602032270
0101 = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x5930 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
▼ [Timestamps]
[Time since first frame in this TCP stream: 169.607500 milliseconds]
[Time since previous frame in this TCP stream: 660.000 microseconds]
> [SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

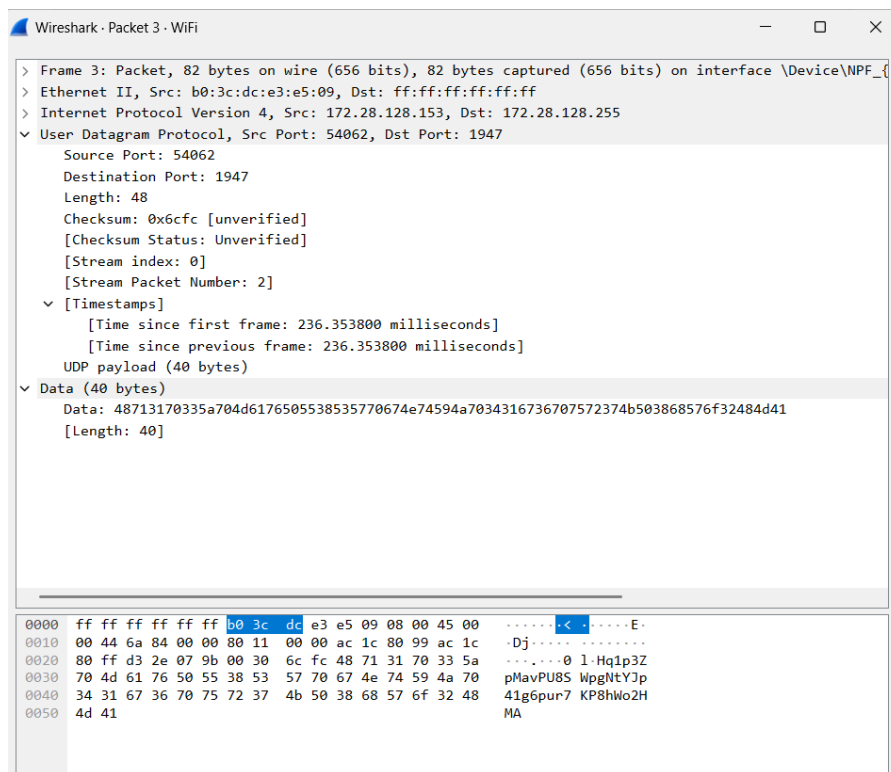
0000 fe ea 05 6c 40 99 b0 3c dc e3 e5 09 08 00 45 00 ...l@<.....E..
0010 00 28 c1 1a 40 00 80 06 00 00 ac 1c 80 99 ac 1c ..(:=@@:.)...C..
0020 80 43 ee 57 00 35 96 7a 66 7c d6 b2 a6 8e 50 10 ...C.W.5.z f|....P..
0030 00 ff 59 30 00 00 ...Y0...

No: 107 - Time: 1.332980400 - Source: 172.28.128.153 - Destination: 172.28.128.67 - Info: 61015 → 53 [ACK] Seq=45 Ack=132 Win=65280 Len=0

Show packet bytes Layout: Vertical (Stacked) ▼

Close Help

UDP



3. Explore traceroute/tracert for different websites eg:google.com and analyse the parameters in the output and explore different options for traceroute command

The tracert command helps identify the path taken by packets across a network and diagnose latency or connectivity issues. By analyzing hop count, delay, and timeouts, network performance and routing problems can be understood.

- Initial hops belong to local network (router, ISP)
- Middle hops represent intermediate ISP routers
- Final hop is destination server
- Delay increases as distance increases
- Uses ICMP Echo Requests
- Sends packets with increasing TTL (Time To Live)
- Each router decreases TTL by 1
- When TTL = 0, router sends back a response
- This helps identify each hop in the path

```

Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yasha>3. Explore traceroute/tracert for different websites eg:google.com and a
ut and explore different options for traceroute command
'3.' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\yasha>tracert google.com

Tracing route to google.com [2404:6800:4007:832::200e]
over a maximum of 30 hops:
  1    3 ms    2 ms    1 ms    2409:40f4:a5:aa2d::fe
  2    *      *      *      Request timed out.
  3   247 ms   13 ms   17 ms   2405:200:5218:21:3925::1
  4   238 ms   13 ms   17 ms   2405:200:88c:1513:62::6
  5    *      *      *      Request timed out.
  6   243 ms   19 ms   17 ms   2405:200:801:900::1626
  7    *      *      *      Request timed out.
  8   244 ms   16 ms   78 ms   2001:4860:1:1::170
  9   227 ms   19 ms   20 ms   2001:4860:1:1::170
 10   40 ms    72 ms   243 ms   2404:6800:8202:280::1
 11   135 ms   232 ms   244 ms   2001:4860:0:1::5658
 12   249 ms   224 ms   238 ms   2001:4860:0:1::34c9
 13   252 ms   17 ms   18 ms   pnmaaa-az-in-x0e.1e100.net [2404:6800:4007:832::200e]

Trace complete

```

Parameters in Output

- Hop Number (1, 2, 3, ...)
Each hop represents a router between source and destination.
- Round Trip Time (ms)
Three time values are shown for each hop, indicating delay for three probe packets.
- IP Address / Hostname
Shows the router or device at that hop.
- * (Asterisk)
Indicates timeout or no response from that hop.

Common tracert Options

-d (Do not resolve hostnames)
tracert -d google.com

- Faster output (skips DNS lookup)

```

C:\Users\yasha>tracert -d google.com

Tracing route to google.com [2404:6800:4007:832::200e]
over a maximum of 30 hops:
  1    2 ms    3 ms    2 ms    2409:40f4:a5:aa2d::fe
  2    *      *      *      Request timed out.
  3   61 ms   27 ms   19 ms   2405:200:5218:21:3925::1
  4   41 ms   18 ms   17 ms   2405:200:88c:1513:62::6
  5    *      *      *      Request timed out.
  6   150 ms   13 ms   13 ms   2405:200:801:900::1626
  7    *      *      *      Request timed out.
  8   261 ms   35 ms   287 ms   2001:4860:1:1::170
  9   250 ms   21 ms   28 ms   2001:4860:1:1::170
 10   46 ms    74 ms   17 ms   2404:6800:8202:280::1
 11   239 ms   13 ms   18 ms   2001:4860:0:1::5658
 12   46 ms   23 ms   15 ms   2001:4860:0:1::34c9
 13   246 ms   11 ms   20 ms   2404:6800:4007:832::200e

```

-h (Maximum hops)

tracert -h 10 google.com

- Limits number of hops to 10

```
C:\Users\yasha>tracert -h 10 google.com

Tracing route to google.com [2404:6800:4007:836::200e]
over a maximum of 10 hops:

  1    1 ms    1 ms    1 ms    2409:40f4:a5:aa2d::fe
  2    *      *      *      Request timed out.
  3   42 ms   27 ms   13 ms   2405:200:5218:21:3925::1
  4   43 ms   18 ms   18 ms   2405:200:88c:1513:62::6
  5    *      *      *      Request timed out.
  6  236 ms   14 ms   15 ms   2405:200:801:900::1628
  7    *      *      *      Request timed out.
  8   79 ms   35 ms   32 ms   2001:4860:1:1::170
  9   45 ms   15 ms   236 ms  2001:4860:1:1::170
 10   21 ms   15 ms   24 ms   2404:6800:8201:380::1

Trace complete.
```

-w (Timeout)

tracert -w 1000 google.com

- Wait time in milliseconds for each reply

```
C:\Users\yasha>tracert -w 1000 google.com

Tracing route to google.com [2404:6800:4007:836::200e]
over a maximum of 30 hops:

  1   222 ms    2 ms    3 ms    2409:40f4:a5:aa2d::fe
  2    *      *      *      Request timed out.
  3  243 ms   28 ms   17 ms   2405:200:5218:21:3925::1
  4  245 ms   11 ms   15 ms   2405:200:88c:1513:62::6
  5    *      *      *      Request timed out.
  6   62 ms   21 ms   23 ms   2405:200:801:900::1628
  7    *      *      *      Request timed out.
  8  228 ms   217 ms  243 ms   2001:4860:1:1::170
  9  200 ms   223 ms  228 ms   2001:4860:1:1::170
 10  245 ms   17 ms   238 ms   2404:6800:8201:380::1
 11  104 ms   22 ms   13 ms   2001:4860:0:1::559a
 12   22 ms   37 ms   28 ms   2001:4860:0:1::55d3
 13  235 ms   195 ms  224 ms   lcmaaa-az-in-x0e.1e100.net [2404:6800:4007:836::200e]

Trace complete.
```

-4 (IPv4)

tracert -4 [google.com](https://www.google.com)

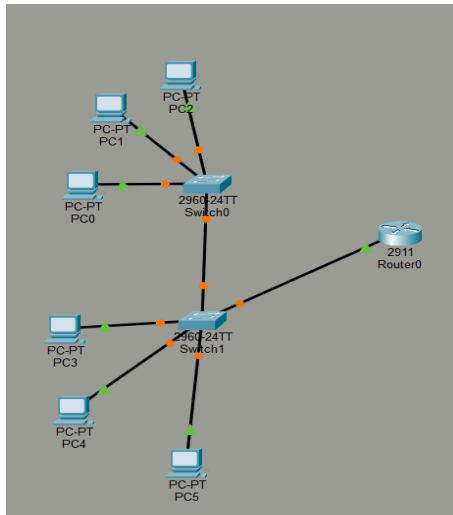
```
C:\Users\yasha>tracert -4 google.com

Tracing route to google.com [142.251.221.110]
over a maximum of 30 hops:

  1    1 ms    1 ms    2 ms   172.28.128.67
  2    *      *      *      Request timed out.
  3    *      *      *      Request timed out.
  4    *      *      *      Request timed out.
  5  179 ms   18 ms   251 ms   172.27.251.98
  6  240 ms   19 ms   18 ms   192.168.142.20
  7    *      *      *      Request timed out.
  8    *      *      *      Request timed out.
  9   46 ms   23 ms   37 ms   209.85.175.48
 10   46 ms   18 ms   16 ms   216.239.51.91
 11  257 ms   27 ms   224 ms   142.251.55.69
 12  179 ms   22 ms   25 ms   cgk03s03-in-f14.1e100.net [142.251.221.110]

Trace complete.
```

4. Set up trunk ports between switches and try pinging between different vlans



vlan configuration on switch0

```
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	10	a-full	a-100	10/100BaseTX
Fa0/2		connected	10	a-full	a-100	10/100BaseTX
Fa0/3		connected	10	a-full	a-100	10/100BaseTX
Fa0/4		connected	1	a-full	a-100	10/100BaseTX

```
Switch>enable
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

Setting up the trunk port

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport mode trunk
```

```
Fa0/24          connected    trunk      a-full  a-100 10/100BaseTX
```

Vlan configuration on switch1

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
20	Fin	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
```

```
Fa0/24          connected    trunk      a-full  a-100 10/100BaseTX
```

Enabling the interface in router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset  up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#
```

Router on stick architecture configuring the interfaces for vlan

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

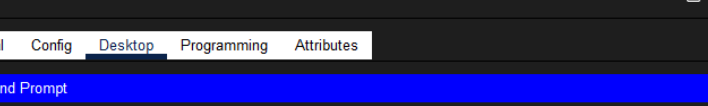
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to
up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

On switch1 shows the trunked ports and vlans allowed on those trunked ports

```
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#  
Switch#show interfaces trunk  
  
Port      Mode      Encapsulation  Status      Native vlan  
Fa0/1     on        802.1q         trunking    1  
Fa0/24    on        802.1q         trunking    1  
  
Port      Vlans allowed on trunk  
Fa0/1     1-1005  
Fa0/24    1-1005  
  
Port      Vlans allowed and active in management domain  
Fa0/1     1,10,20  
Fa0/24    1,10,20  
  
Port      Vlans in spanning tree forwarding state and not pruned  
Fa0/1     1,10,20  
Fa0/24    1,10,20  
  
Switch#
```

Pinging pcs in two different [vlans. It](#) has to be routed via the routers.



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.20.10

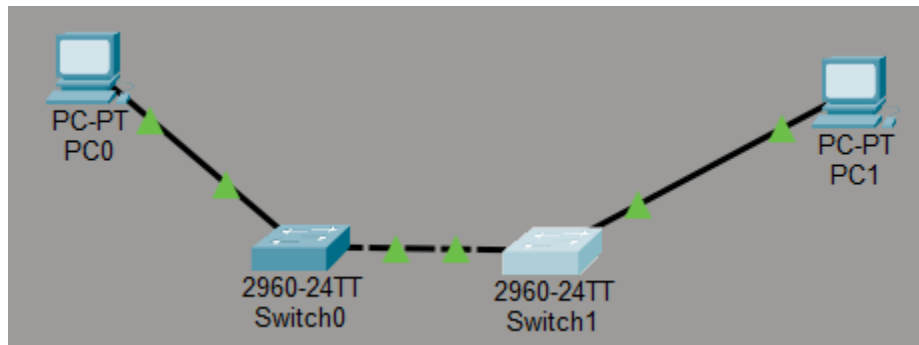
Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5. Change the native VLAN on a trunk port. Test for VLAN mismatches and Troubleshoot.

Network topology



Configuring the vlan. Two types of ports access and trunk ports are being configured.

```
Switch0
Physical Config CLI Attributes
IOS Command

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNT
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN10
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10

Switch(config-if)#interface fa0/5
Switch(config-if)#switchport mode trunk
```

```
Switch0
Physical Config CLI Attributes

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gi0/1, Gi0/2
10   VLAN10                 active    Fa0/1
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

Switch#
Switch#
Switch#show interfaces status

Port    Name        Status    Vlan    Duplex  Speed Type
-----
Fa0/1                connected 10      a-full  a-100  10/100BaseTX
Fa0/2                notconnect 1        auto    auto   10/100BaseTX
Fa0/3                notconnect 1        auto    auto   10/100BaseTX
Fa0/4                notconnect 1        auto    auto   10/100BaseTX
Fa0/5                connected  trunk   a-full  a-100  10/100BaseTX
Fa0/6                notconnect 1        auto    auto   10/100BaseTX
```

```
Switch1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if)#interface fa0/24
Switch(config-if)#interface fa0/5
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   VLAN0010                active    Fa0/1
20   VLAN20                  active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
Switch#
Switch#
Switch#show interfaces status

Port      Name                Status    Vlan    Duplex  Speed Type
-----
Fa0/1     Fa0/1               connected  10      a-full  a-100  10/100BaseTX
Fa0/2     Fa0/2               notconnect 1        auto    auto   10/100BaseTX
Fa0/3     Fa0/3               notconnect 1        auto    auto   10/100BaseTX
Fa0/4     Fa0/4               notconnect 1        auto    auto   10/100BaseTX
Fa0/5     Fa0/5               connected  trunk   a-full  a-100  10/100BaseTX
Fa0/6     Fa0/6               notconnect 1        auto    auto   10/100BaseTX
```

Changing the native vlan on switch0

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/5
Switch(config-if)#switchport trunk native vlan 99
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

The show interfaces trunk command helps us troubleshoot by seeing that the native vlan is different in both the switches

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/5     on        802.1q         trunking    99

Port      Vlans allowed on trunk
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/5     1,10

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/5     1,10

Switch#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (99), with Switch
FastEthernet0/5 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (99), with Switch
FastEthernet0/5 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (99), with Switch
FastEthernet0/5 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (99), with Switch
FastEthernet0/5 (1).
```

```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/5     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/5     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/5     1,10,20

Switch#
Switch#
Switch#
Switch#
Switch#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (1), with Switch FastEthernet0/5 (99).
```

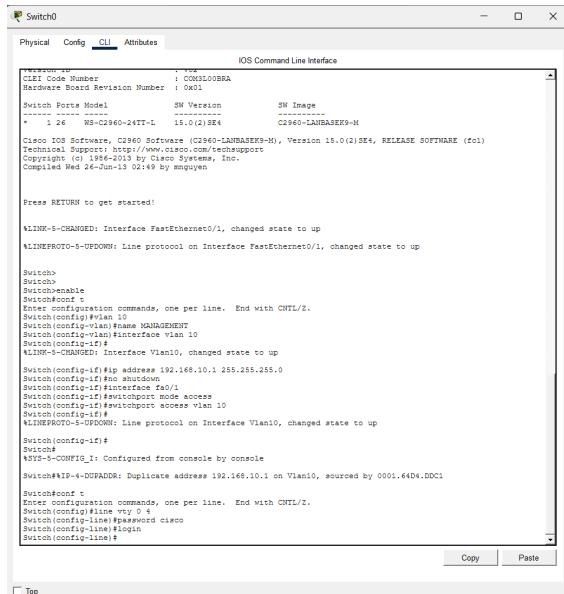
Native VLAN is VLAN for untagged traffic.

Mismatch Causes:

- Traffic leaks
- VLAN hopping risk
- Misdelivery

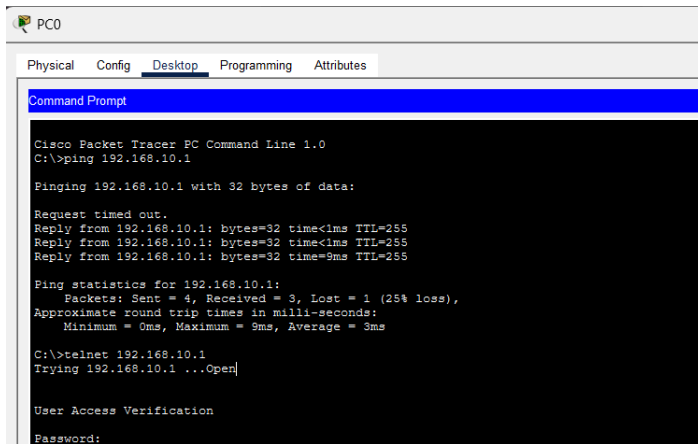
6. Configuring the SVI interface so that the pc can access the switch remotely using telnet or ssh.

Telnet conf



Ssh conf

```
Switch(config)#hostname Switch
Switch(config)#ip domain-name example.com
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#hostname yasha
yasha(config)#ip domain-name example.com
yasha(config)#crypto key generate rsa
The name for the keys will be: yasha.example.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 450
% Generating 450 bit RSA keys, keys will be non-exportable...[OK]
yasha(config)#username admin password cisco
*Mar 1 0:13:5.87: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:13:5.87: %SSH-5-ENABLED: SSH 1.5 has been enabled
yasha(config)#line vty 0 4
yasha(config-line)#login local
yasha(config-line)#transport input ssh
yasha(config-line)#
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

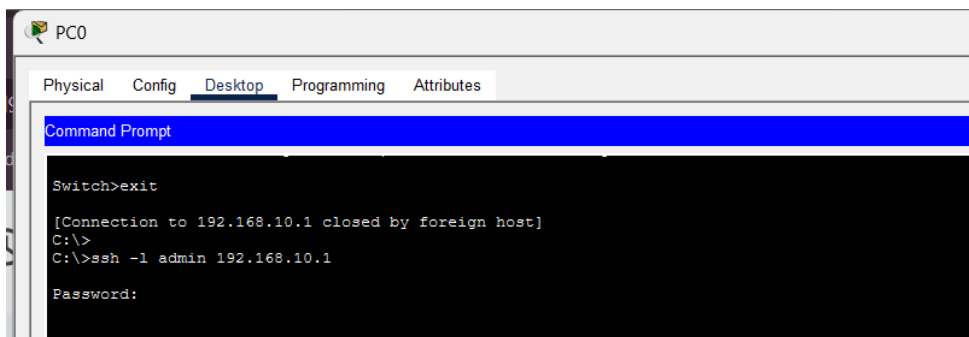
Request timed out.
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=9ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>telnet 192.168.10.1
Trying 192.168.10.1 ...Open

User Access Verification
Password:
```

Closing the telnet connection and trying to check if the switch can be accessed using ssh



```
Switch>exit

[Connection to 192.168.10.1 closed by foreign host]
C:\>
C:\>>ssh -l admin 192.168.10.1

Password:
```

Management VLAN?

A Management VLAN is a VLAN used only to manage the switch remotely (SSH/Telnet).

- Switch is a Layer 2 device, so it normally has NO IP
- To access it remotely → we give an IP to a VLAN interface (SVI)

SVI (Switched Virtual Interface)

Logical interface on a switch. IP addresses are not assigned to physical ports on L2 switches they are assigned to SVI.

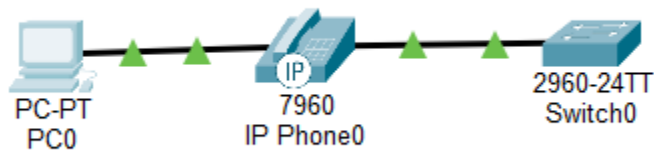
VLAN for management?

- Security (separate traffic)
- Organization
- Remote access (SSH/Telnet)

Telnet is insecure because credentials are sent in plaintext, so SSH is preferred.”

7. You have a Cisco switch and a VoIP phone that needs to be placed in a voice VLAN (VLAN 20). The data for the PC should remain in a separate VLAN (VLAN 10). Configure the switch port to support both voice and data traffic.

Network topology



Configuring the vlans to carry data and voice.

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name DATA
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name VOICE
Switch(config-vlan)#
Switch(config-vlan)#exit

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```



```

Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch#
Switch#
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
10   DATA                  active    Fa0/1
20   VOICE                  active    Fa0/1
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

```

```

Switch#show ip interface brief

Interface      IP-Address      OK? Method Status    Protocol
FastEthernet0/1 unassigned      YES manual up        up
FastEthernet0/2 unassigned      YES manual down    down
FastEthernet0/3 unassigned      YES manual down    down
FastEthernet0/4 unassigned      YES manual down    down
FastEthernet0/5 unassigned      YES manual down    down
FastEthernet0/6 unassigned      YES manual down    down
FastEthernet0/7 unassigned      YES manual down    down
FastEthernet0/8 unassigned      YES manual down    down
FastEthernet0/9 unassigned      YES manual down    down
FastEthernet0/10 unassigned      YES manual down    down
FastEthernet0/11 unassigned      YES manual down    down
FastEthernet0/12 unassigned      YES manual down    down
FastEthernet0/13 unassigned      YES manual down    down
FastEthernet0/14 unassigned      YES manual down    down
FastEthernet0/15 unassigned      YES manual down    down
FastEthernet0/16 unassigned      YES manual down    down
FastEthernet0/17 unassigned      YES manual down    down
FastEthernet0/18 unassigned      YES manual down    down
FastEthernet0/19 unassigned      YES manual down    down
FastEthernet0/20 unassigned      YES manual down    down
FastEthernet0/21 unassigned      YES manual down    down
FastEthernet0/22 unassigned      YES manual down    down
FastEthernet0/23 unassigned      YES manual down    down
FastEthernet0/24 unassigned      YES manual down    down
GigabitEthernet0/1 unassigned      YES manual down    down
GigabitEthernet0/2 unassigned      YES manual down    down
Vlan1          unassigned      YES manual administratively down down
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

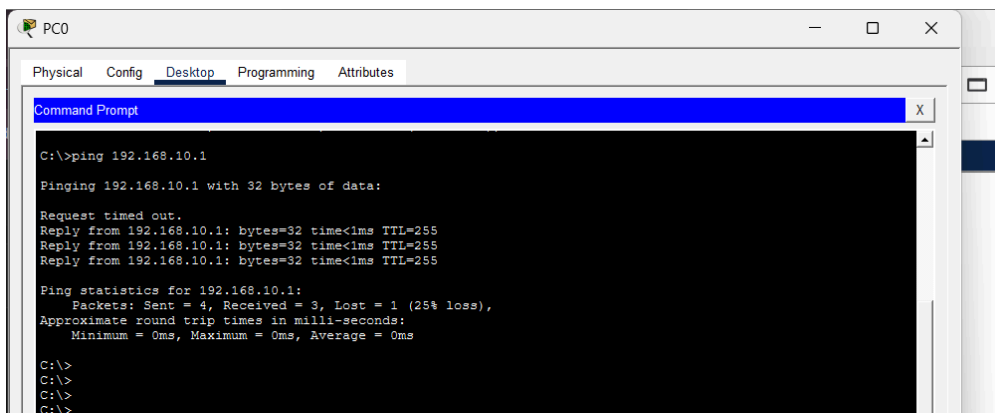
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#

```

This configuration allows both the IP phone and PC to share the same switch port. The IP phone tags voice traffic for VLAN 20, while the PC sends untagged data traffic that is placed in VLAN 10.

8. Configure a Cisco switch port to support both voice (VLAN 20) and data (VLAN 10) traffic, we use a combination of access VLAN and voice VLAN configuration.
- switchport access vlan 10: Assigns untagged data traffic (PC) to VLAN 10
 - switchport voice vlan 20: Assigns tagged voice traffic (IP Phone) to VLAN 20
 - switchport mode access: Configures the port as an access port
 - spanning-tree portfast : Enables fast transition to forwarding state (useful for end devices)

This configuration allows both the IP phone and PC to share the same switch port. The IP phone tags voice traffic for VLAN 20, while the PC sends untagged data traffic that is placed in VLAN 10.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

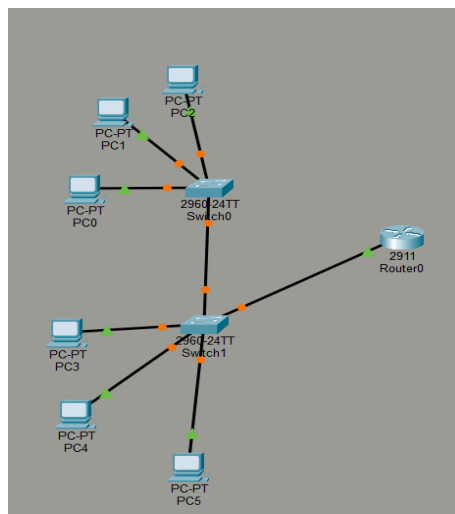
Request timed out.
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>
C:\>
C:\>
```

9. Try Inter VLAN routing with Router.

Set up trunk ports between switches and try pinging between different vlans



vlan configuration on switch0

```
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	10	a-full	a-100	10/100BaseTX
Fa0/2		connected	10	a-full	a-100	10/100BaseTX
Fa0/3		connected	10	a-full	a-100	10/100BaseTX
Fa0/4		connected	1	a-full	a-100	10/100BaseTX

```
Switch>enable
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Setting up the trunk port

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport mode trunk
```

Fa0/24	connected	trunk	a-full	a-100	10/100BaseTX
--------	-----------	-------	--------	-------	--------------

Vlan configuration on switch1

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
20	Fin	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

```
Switch>enable
```

```
Switch#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Switch(config)#interface Fa0/24
```

```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#exit
```

```
Switch(config)#
```

```
Switch(config)#exit
```

```
| Fa0/24 connected trunk a-full a-100 10/100BaseTX
```

Enabling the interface in router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset up up
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

Router on stick architecture configuring the interfaces for vlan

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

```
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/24    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/24    1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20
Fa0/24    1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20
Fa0/24    1,10,20

Switch#
```



The screenshot shows a Windows PC0 interface with a dark theme. The top bar displays 'PC0' and standard window controls. Below the bar are tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The command prompt has a blue title bar and a black background. The command 'C:\>ping 192.168.20.10' has been entered and executed. The output shows a successful ping with 32 bytes of data, a request time of 0ms, and a TTL of 127. The statistics show 4 packets sent, 3 received, and 1 lost (25% loss).

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
show vlan brief
```

- Ensure VLAN 10 and VLAN 20 exist
- Ensure correct ports are assigned

Step 2: Check Interface Status

show ip interface brief

- Ensure interfaces are up/up

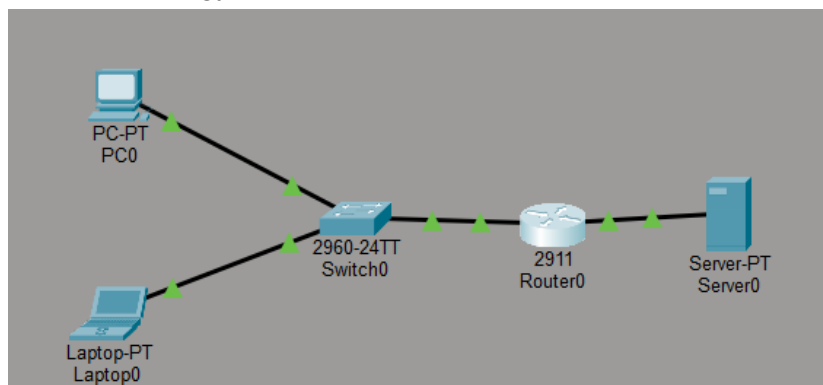
Step 3: Identify Root Cause

- VLANs are isolated at Layer 2
- No routing device is configured
- Therefore, communication between VLANs is not possible

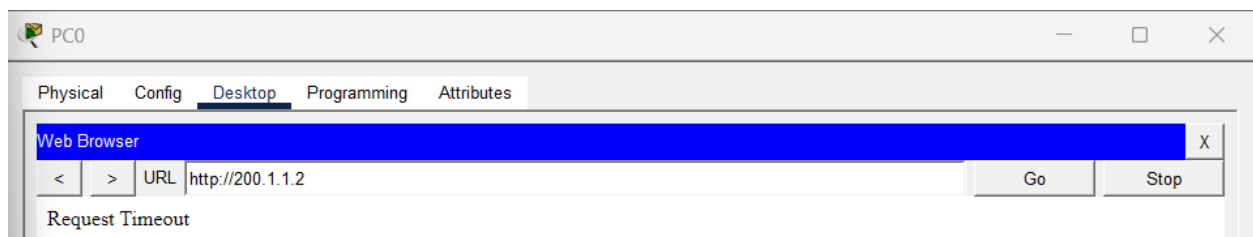
Step 4: Solution – Enable Inter-VLAN Routing

11. Implement ACLs to restrict traffic based on source and destination ports. Test rules by simulating legitimate and unauthorized traffic.

Network topology



FTP and HTTP services fail.



```
C:\>ftp 200.1.1.2
Trying to connect...200.1.1.2

%Error opening ftp://200.1.1.2/ (Timed out)
.

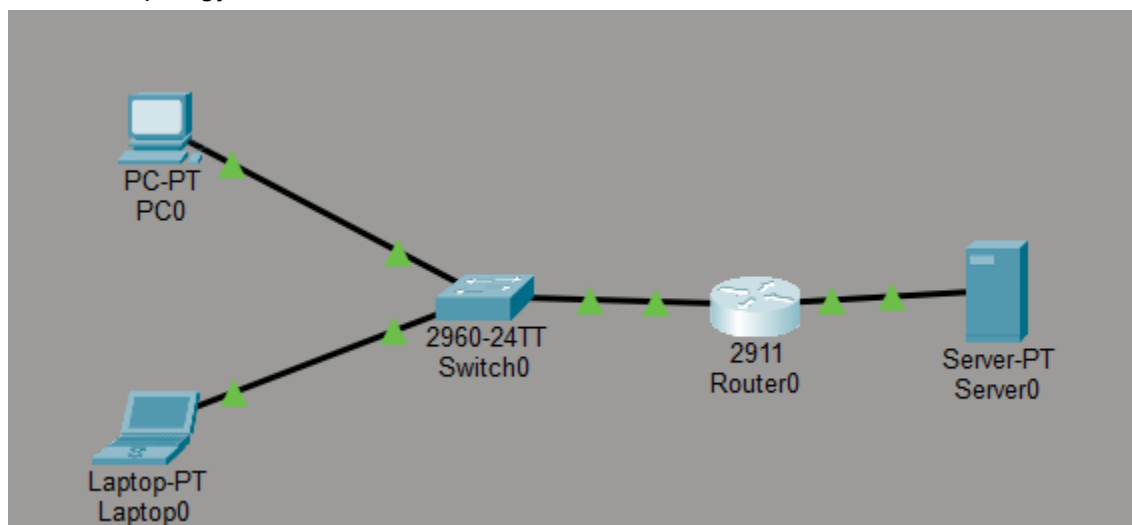
(Disconnecting from ftp server)
```

blocking HTTP and FTP services is an example of destination port-based filtering, as it restricts access to services running on specific ports. Source port filtering can also be applied to restrict traffic originating from specific services.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny tcp any eq 80 any
Router(config)#access-list 100 deny tcp any any eq 80
Router(config)#
```

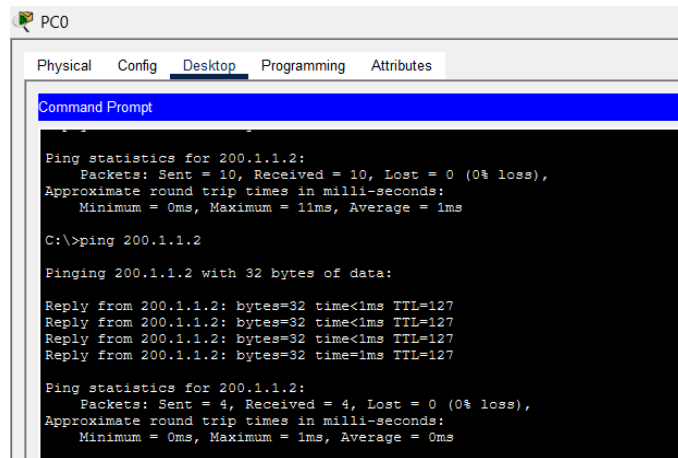
12. Configure a standard Access Control List (ACL) on a router to permit traffic from a specific IP range. Test connectivity to verify the ACL is working as intended.

Network topology



Pinging from pc0 works because the ACL rule allowed to traffic from this ip.

Pinging from laptop0 doesnt work because the ACL rule did not allow to traffic from this ip.



PC0

Physical Config Desktop Programming Attributes

Command Prompt

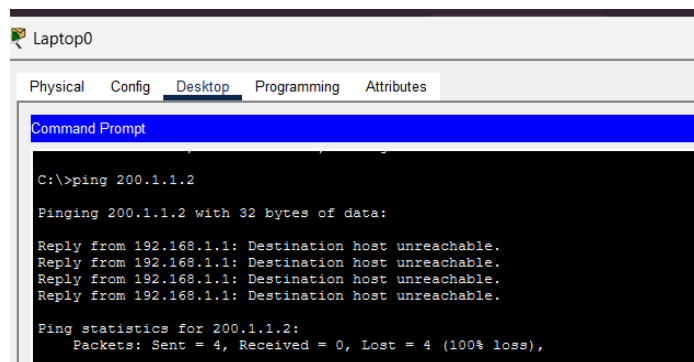
```
Ping statistics for 200.1.1.2:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 1ms

C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



Laptop0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Configuring the acl rules

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 10 permit 192.168.1.2
Router(config)#access-list 10 deny any
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-list
Standard IP access list 10
 10 permit host 192.168.1.2
 20 deny any

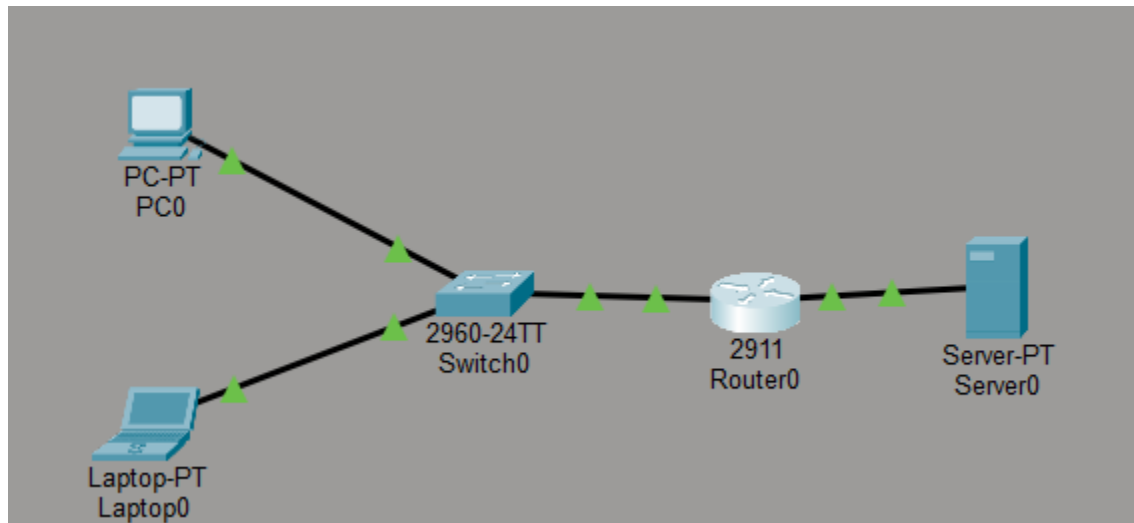
Router#ip interface g0/2
^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/2
Router(config-if)#ip access-group 10 out
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

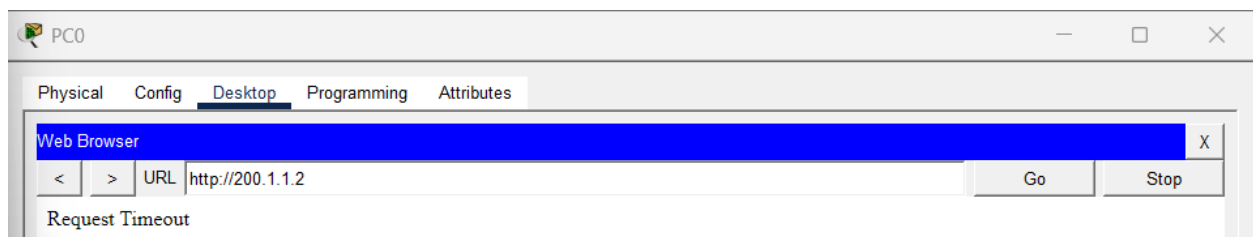

13. Create an extended ACL to block specific applications, such as HTTP or FTP traffic. Test the ACL rules by attempting to access blocked services.

Network topology



```
Router(config)#  
Router(config)#  
Router(config)#access-list 100 deny tcp any any eq 80  
Router(config)#access-list 100 deny tcp any any eq 21  
Router(config)#access-list 100 permit ip any any  
Router(config)#interface g0/1  
Router(config-if)#ip access-group 100 in  
Router(config-if)#
```

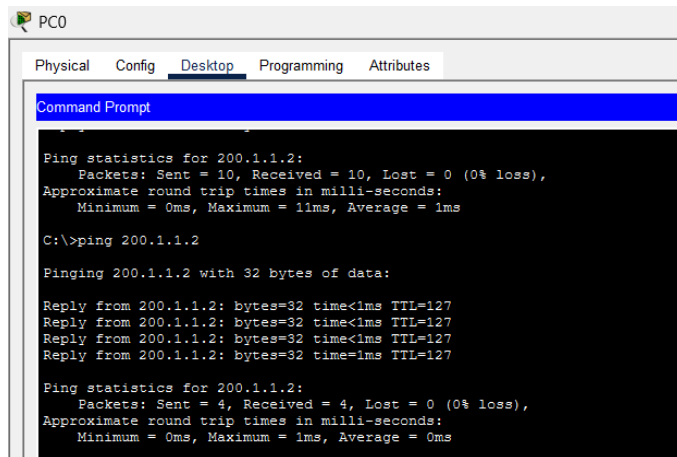
FTP and HTTP services fail but ping works because the ping traffic is allowed by the ACL.



```
C:\>ftp 200.1.1.2
Trying to connect...200.1.1.2

%Error opening ftp://200.1.1.2/ (Timed out)
.

(Disconnecting from ftp server)
```



The screenshot shows a PC0 desktop environment with a Command Prompt window open. The window displays the results of a ping command to 200.1.1.2. The output shows that the first 10 packets were sent and received with 0% loss, and the round trip times were approximately 0ms to 1ms. The second 4 packets were also sent and received with 0% loss, and the round trip times were approximately 0ms to 1ms.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Ping statistics for 200.1.1.2:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 1ms

C:\>ping 200.1.1.2

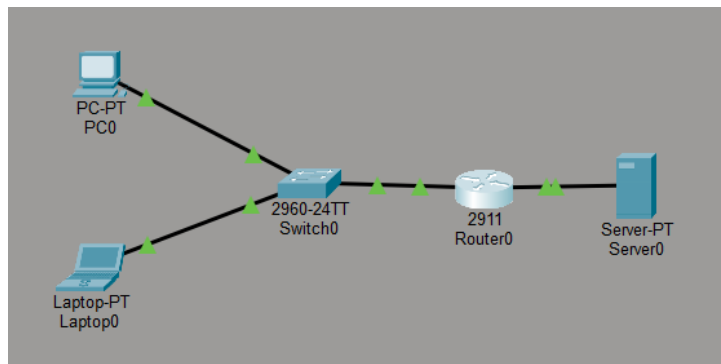
Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

14. Try Static NAT, Dynamic NAT and PAT to translate IPs

Network topology



IP addresses of the end hosts. The default gateway is changed from 0.0.0.0 to 192.168.1.1.

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

Laptop0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static This address is already used in the network.

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

Server0

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 200.1.1.2

Subnet Mask 255.255.255.0

Default Gateway 200.1.1.1

DNS Server 0.0.0.0

Configuring the interfaces one for the LAN side another for the WAN side

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/2
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
```

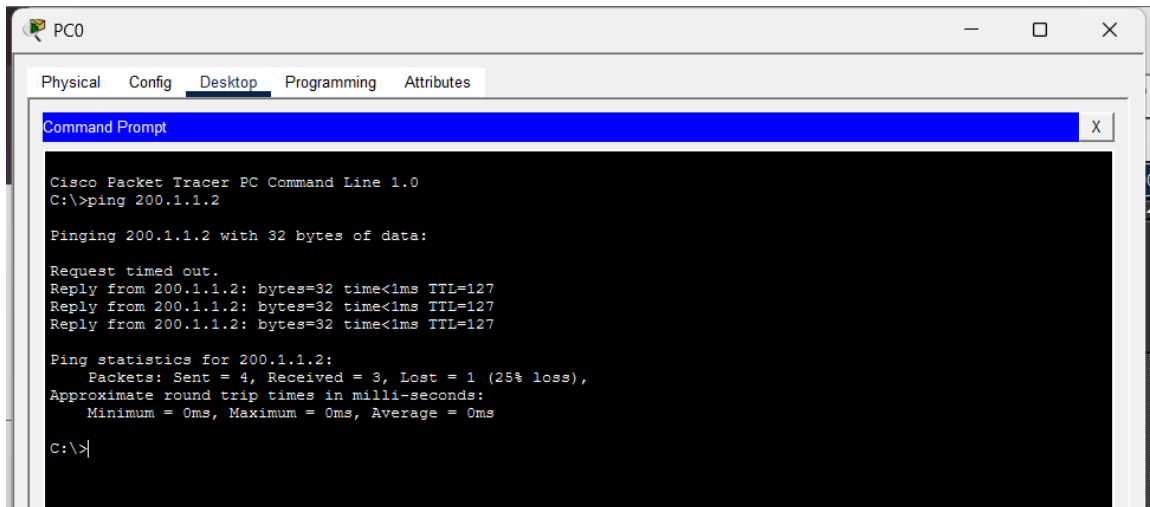
Shows the interfaces and the routes

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
    200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/24 is directly connected, GigabitEthernet0/2
L       200.1.1.1/32 is directly connected, GigabitEthernet0/2
```

Pinging the server before configuring NAT. Before configuring NAT, communication between internal and external network was possible because both networks were directly connected to the router and present in the routing table. NAT is required only when translating private IP addresses for communication with external networks such as the Internet, where private IPs are not routable.



STATIC NAT

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip nat inside source static 192.168.1.2 200.1.1.100
Router(config)#interface g0/1
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface g0/2
Router(config-if)#ip nat outside
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
```

NAT translations

```
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.100:21    192.168.1.2:21    200.1.1.2:21      200.1.1.2:21
icmp 200.1.1.100:22    192.168.1.2:22    200.1.1.2:22      200.1.1.2:22
icmp 200.1.1.100:23    192.168.1.2:23    200.1.1.2:23      200.1.1.2:23
icmp 200.1.1.100:24    192.168.1.2:24    200.1.1.2:24      200.1.1.2:24
--- 200.1.1.100        192.168.1.2      ---                ---

Router#
```

```

C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

DYNAMIC NAT

```

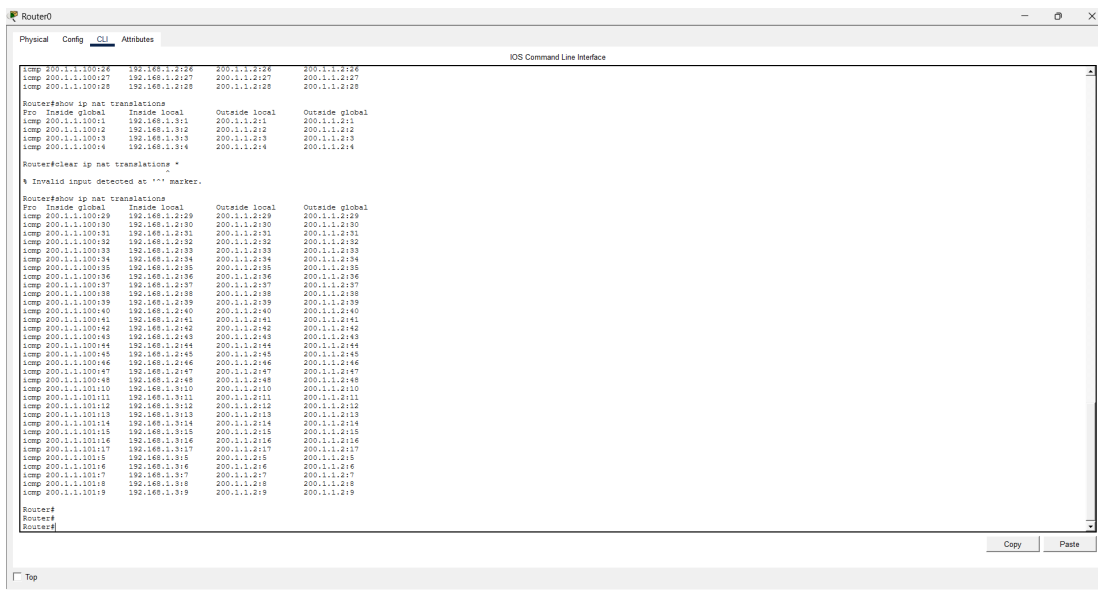
Router(config)#no ip nat inside source static 192.168.1.2 200.1.1.100
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip nat pool MYPOOL 200.1.1.100 200.1.1.110 netmask 255.255.255.0
Router(config)#access-list permit 192.168.1.0 0.0.0.255
^
% Invalid input detected at '^' marker.

Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 pool MYPOOL
Router(config)#exit
Router#

```

Each private ip gets mapped to a public ip.



[illegible]

```
Laptop0
```

Physical Config Desktop Networking Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 200.1.1.2 -n 20

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.1:2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 20, Received = 20, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PAT
Many private ips share one public ip.

```
Router(config)#ip nat inside source list 1 interface g0/2 overload
Router(config)#exit
Router#
*SYS-5-CONFIG I: Configured from console by console
show ip nat translations
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.1.1:45       192.168.1.3:45   200.1.1.2:45      200.1.1.2:45
icmp 200.1.1.1.1:46   192.168.1.3:46   200.1.1.2:46      200.1.1.2:46
icmp 200.1.1.1.1:47   192.168.1.3:47   200.1.1.2:47      200.1.1.2:47
icmp 200.1.1.1.1:48   192.168.1.3:48   200.1.1.2:48      200.1.1.2:48
icmp 200.1.1.1.1:49   192.168.1.3:49   200.1.1.2:49      200.1.1.2:49
icmp 200.1.1.1.1:50   192.168.1.3:50   200.1.1.2:50      200.1.1.2:50
icmp 200.1.1.1.1:51   192.168.1.3:51   200.1.1.2:51      200.1.1.2:51
icmp 200.1.1.1.1:52   192.168.1.3:52   200.1.1.2:52      200.1.1.2:52
icmp 200.1.1.1.1:53   192.168.1.3:53   200.1.1.2:53      200.1.1.2:53
icmp 200.1.1.1.1:54   192.168.1.3:54   200.1.1.2:54      200.1.1.2:54
icmp 200.1.1.1.1:69   192.168.1.2:69   200.1.1.2:69      200.1.1.2:69
icmp 200.1.1.1.1:70   192.168.1.2:70   200.1.1.2:70      200.1.1.2:70
icmp 200.1.1.1.1:71   192.168.1.2:71   200.1.1.2:71      200.1.1.2:71
icmp 200.1.1.1.1:72   192.168.1.2:72   200.1.1.2:72      200.1.1.2:72
icmp 200.1.1.1.1:73   192.168.1.2:73   200.1.1.2:73      200.1.1.2:73
icmp 200.1.1.1.1:74   192.168.1.2:74   200.1.1.2:74      200.1.1.2:74
icmp 200.1.1.1.1:75   192.168.1.2:75   200.1.1.2:75      200.1.1.2:75
icmp 200.1.1.1.1:76   192.168.1.2:76   200.1.1.2:76      200.1.1.2:76
icmp 200.1.1.1.1:77   192.168.1.2:77   200.1.1.2:77      200.1.1.2:77
icmp 200.1.1.1.1:78   192.168.1.2:78   200.1.1.2:78      200.1.1.2:78

Router#
```

15. Download iperf in laptop/phone and make sure they are in same network. Try different iperf commands with tcp, udp, bidirectional, reverse, multicast, parallel options and analyze the bandwidth and rate of transmission, delay, jitter etc.

Iperf3 has to be downloaded as a zip file and extracted.

```
PS E:\> cd Downloads
PS E:\Downloads> cd "iperf3.1.4_64"
PS E:\Downloads\iperf3.1.4_64> ls

Directory: E:\Downloads\iperf3.1.4_64

Mode                LastWriteTime         Length Name
----                -
-a----             22-03-2026     10:00       3306962 cygwin1.dll
-a----             22-03-2026     10:00       472387   iperf3.exe
```

```
Administrator: Windows PowerShell
PS E:\Downloads\iperf3.1.4_64> iperf3.exe
iperf3.exe : The term 'iperf3.exe' is not recognized as the name of a cmdlet, function, script file,
or operable program. Check the spelling of the name, or if a path was included, verify that the path
is correct and try again.
At line:1 char:1
+ iperf3.exe
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (iperf3.exe:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

Suggestion [3,General]: The command iperf3.exe was not found, but does exist in the current location. Wi
ndows PowerShell does not load commands from the current location by default. If you trust this command,
instead type: ".\iperf3.exe". See "get-help about_Command_Precedence" for more details.
PS E:\Downloads\iperf3.1.4_64> .\iperf3.exe
iperf3: parameter error - must either be a client (-c) or server (-s)

Usage: iperf [-s|-c host] [options]
       iperf [-h|--help] [-v|--version]

Server or Client:
-p, --port #                server port to listen on/connect to
-f, --format [kmgKMG]       format to report: Kbits, Mbits, KBytes, MBytes
-i, --interval #            seconds between periodic bandwidth reports
-F, --file name             xmit/rcv the specified file
-B, --bind <host>          bind to a specific interface
-V, --verbose               more detailed output
-J, --json                  output in JSON format
--logfile f                 send output to a log file
-d, --debug                 emit debugging output
-v, --version               show version information and quit
-h, --help                  show this message and quit

Server specific:
-s, --server                run in server mode
-D, --daemon                run the server as a daemon
-I, --pidfile file          write PID file
-l, --one-off                handle one client connection then exit

Client specific:
-c, --client <host>        run in client mode, connecting to <host>
-u, --udp                   use UDP rather than TCP
-b, --bandwidth #[KMG]/[#] target bandwidth in bits/sec (0 for unlimited)
                           (default 1 Mbit/sec for UDP, unlimited for TCP)
                           (optional slash and packet count for burst mode)
-t, --time #                time in seconds to transmit for (default 10 secs)
-n, --bytes #[KMG]          number of bytes to transmit (instead of -t)
-k, --blockcount #[KMG]     number of blocks (packets) to transmit (instead of -t or -n)
-l, --len #[KMG]            length of buffer to read or write
                           (default 128 KB for TCP, 8 KB for UDP)
--cport <port>              bind to a specific client port (TCP and UDP, default: ephemeral port)
```

The laptop and phone need to be in the same network. The laptop acts as the server and the phone acts as the client.

The server is started on the laptop and the iperf3 on the phone must be configured with the corresponding ip address to see the connectivity metrics.

```
PS E:\Downloads\iperf3.1.4_64> .\iperf3 -s
-----
Server listening on 5201
-----
Accepted connection from 172.28.128.67, port 53534
[ 5] local 172.28.128.153 port 5201 connected to 172.28.128.67 port 53540
[ ID] Interval      Transfer    Bandwidth
[ 5]  0.00-1.01  sec   3.62 MBytes  30.1 Mbits/sec
[ 5]  1.01-2.01  sec   2.12 MBytes  17.8 Mbits/sec
[ 5]  2.01-3.00  sec   5.88 MBytes  49.7 Mbits/sec
[ 5]  3.00-4.00  sec   6.88 MBytes  57.5 Mbits/sec
[ 5]  4.00-5.00  sec   1.75 MBytes  14.7 Mbits/sec
[ 5]  5.00-6.02  sec   3.62 MBytes  30.0 Mbits/sec
[ 5]  6.02-7.00  sec   4.62 MBytes  39.4 Mbits/sec
[ 5]  7.00-8.00  sec   7.62 MBytes  63.9 Mbits/sec
[ 5]  8.00-9.01  sec   5.38 MBytes  44.6 Mbits/sec
[ 5]  9.01-10.00 sec  11.0 MBytes  93.2 Mbits/sec
[ 5] 10.00-10.01 sec   128 KBytes   114 Mbits/sec
-----
[ ID] Interval      Transfer    Bandwidth
[ 5]  0.00-10.01  sec  52.6 MBytes  44.1 Mbits/sec
[ 5]  0.00-10.01  sec    0.00 Bytes    0.00 bits/sec
-----
Server listening on 5201
-----
```

Using the the same system (laptop) to test out the metrics while the UDP protocol is used.

```
PS E:\Downloads\iperf3.1.4_64> .\iperf3 -s
-----
Server listening on 5201
-----
Accepted connection from ::1, port 57683
[ 5] local ::1 port 5201 connected to ::1 port 62498
[ ID] Interval      Transfer    Bandwidth    Jitter    Lost/Total Datagrams
[ 5]  0.00-1.01  sec   120 KBytes   978 Kbits/sec  0.073 ms  0/15 (0%)
[ 5]  1.01-2.02  sec   128 KBytes   1.04 Mbits/sec  0.060 ms  0/16 (0%)
[ 5]  2.02-3.01  sec   128 KBytes   1.06 Mbits/sec  0.067 ms  0/16 (0%)
[ 5]  3.01-4.00  sec   128 KBytes   1.05 Mbits/sec  0.075 ms  0/16 (0%)
[ 5]  4.00-5.01  sec   128 KBytes   1.04 Mbits/sec  0.119 ms  0/16 (0%)
[ 5]  5.01-6.01  sec   128 KBytes   1.06 Mbits/sec  0.105 ms  0/16 (0%)
[ 5]  6.01-7.01  sec   128 KBytes   1.05 Mbits/sec  0.084 ms  0/16 (0%)
[ 5]  7.01-8.00  sec   128 KBytes   1.06 Mbits/sec  0.123 ms  0/16 (0%)
[ 5]  8.00-9.01  sec   128 KBytes   1.04 Mbits/sec  0.096 ms  0/16 (0%)
[ 5]  9.01-10.01 sec   128 KBytes   1.04 Mbits/sec  0.080 ms  0/16 (0%)
[ 5] 10.01-10.02 sec    0.00 Bytes    0.00 bits/sec  0.080 ms  0/0 (0%)
-----
[ ID] Interval      Transfer    Bandwidth    Jitter    Lost/Total Datagrams
[ 5]  0.00-10.02  sec    0.00 Bytes    0.00 bits/sec  0.080 ms  0/159 (0%)
-----
Server listening on 5201
-----
```

```

PS E:\Downloads\iperf3.1.4.64> .\iperf3 -c localhost -u
Connecting to host localhost, port 5201
[ 4] local ::1 port 62498 connected to ::1 port 5201
[ ID] Interval      Transfer    Bandwidth      Total Datagrams
[ 4]  0.00-1.01    sec      120 KBytes    978 Kbits/sec    15
[ 4]  1.01-2.01    sec      128 KBytes    1.04 Mbits/sec   16
[ 4]  2.01-3.01    sec      128 KBytes    1.06 Mbits/sec   16
[ 4]  3.01-4.00    sec      128 KBytes    1.05 Mbits/sec   16
[ 4]  4.00-5.01    sec      128 KBytes    1.04 Mbits/sec   16
[ 4]  5.01-6.01    sec      128 KBytes    1.06 Mbits/sec   16
[ 4]  6.01-7.01    sec      128 KBytes    1.04 Mbits/sec   16
[ 4]  7.01-8.00    sec      128 KBytes    1.06 Mbits/sec   16
[ 4]  8.00-9.01    sec      128 KBytes    1.04 Mbits/sec   16
[ 4]  9.01-10.01   sec      128 KBytes    1.04 Mbits/sec   16
-----
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 4]  0.00-10.01   sec      1.24 MBytes    1.04 Mbits/sec  0.080 ms   0/159 (0%)
[ 4] Sent 159 datagrams

iperf Done.

```

Setting the bandwidth to 50M

```

-----
Server listening on 5201
-----
Accepted connection from ::1, port 53839
[ 5] local ::1 port 5201 connected to ::1 port 62498
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 5]  0.00-1.01    sec      5.48 MBytes    45.5 Mbits/sec  0.009 ms   0/701 (0%)
[ 5]  1.01-2.01    sec      5.97 MBytes    50.0 Mbits/sec  0.012 ms   1/765 (0.13%)
[ 5]  2.01-3.01    sec      5.96 MBytes    50.2 Mbits/sec  0.009 ms   0/763 (0%)
[ 5]  3.01-4.01    sec      5.94 MBytes    49.7 Mbits/sec  0.010 ms   0/760 (0%)
[ 5]  4.01-5.00    sec      5.96 MBytes    50.1 Mbits/sec  0.011 ms   0/763 (0%)
[ 5]  5.00-6.00    sec      5.91 MBytes    49.8 Mbits/sec  0.019 ms   0/757 (0%)
[ 5]  6.00-7.01    sec      6.03 MBytes    50.0 Mbits/sec  0.009 ms   0/772 (0%)
[ 5]  7.01-8.01    sec      5.88 MBytes    49.3 Mbits/sec  0.019 ms   0/753 (0%)
[ 5]  8.01-9.01    sec      6.02 MBytes    50.9 Mbits/sec  0.011 ms   0/770 (0%)
[ 5]  9.01-10.01   sec      5.95 MBytes    49.8 Mbits/sec  0.019 ms   8/770 (1%)
[ 5] 10.01-10.01   sec       0.00 Bytes     0.00 bits/sec   0.019 ms   0/0 (0%)
-----
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 5]  0.00-10.01   sec       0.00 Bytes     0.00 bits/sec   0.019 ms  9/7574 (0.12%)
-----
Server listening on 5201
-----

```

```

PS E:\Downloads\iperf3.1.4_64> .\iperf3 -c localhost -u -b 50M
Connecting to host localhost, port 5201
[ 4] local ::1 port 62498 connected to ::1 port 5201
[ ID] Interval           Transfer     Bandwidth       Total Datagrams
[ 4]  0.00-1.01   sec  5.48 MBytes  45.7 Mbits/sec   701
[ 4]  1.01-2.01   sec  5.98 MBytes  50.1 Mbits/sec   765
[ 4]  2.01-3.00   sec  5.96 MBytes  50.2 Mbits/sec   763
[ 4]  3.00-4.00   sec  5.94 MBytes  49.7 Mbits/sec   760
[ 4]  4.00-5.00   sec  5.96 MBytes  50.1 Mbits/sec   763
[ 4]  5.00-6.01   sec  5.91 MBytes  49.0 Mbits/sec   757
[ 4]  6.01-7.01   sec  6.03 MBytes  50.8 Mbits/sec   772
[ 4]  7.01-8.01   sec  5.88 MBytes  49.3 Mbits/sec   753
[ 4]  8.01-9.00   sec  6.02 MBytes  50.9 Mbits/sec   770
[ 4]  9.00-10.01  sec  6.02 MBytes  50.3 Mbits/sec   770
- - - - -
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[ 4]  0.00-10.01  sec  59.2 MBytes  49.6 Mbits/sec  0.019 ms  9/7574 (0.12%)
[ 4] Sent 7574 datagrams

iperf Done.
PS E:\Downloads\iperf3.1.4_64>

```